

Fabian Gittins — Publications

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PUBLICATION SUMMARY	Full list of publications can be found on Google Scholar , INSPIRE-HEP and NASA ADS . h-index —15 (according to Google Scholar) or 14 (INSPIRE-HEP). Top five cited —Excluding long-author-list papers. Citation counts from Google Scholar. 1. Gittins, F., Andersson, N., Jones, D. I., <i>Modelling neutron star mountains</i>, Mon. Not. R. Astron. Soc. 500, 5570 (2021) [arXiv:2009.12794]. (80 citations) 2. Gittins, F., Andersson, N., <i>Modelling neutron star mountains in relativity</i>, Mon. Not. R. Astron. Soc. 507, 116 (2021) [arXiv:2105.06493]. (67 citations) 3. Gittins, F., Andersson, N., Pereira, J. P., <i>Tidal deformations of neutron stars with elastic crusts</i>, Phys. Rev. D 101, 103025 (2020) [arXiv:2003.05449]. (48 citations) 4. Pereira, J. P., Bejger, M., Andersson, N., Gittins, F., <i>Tidal deformations of hybrid stars with sharp phase transitions and elastic crusts</i>, Astrophys. J. 895, 28 (2020) [arXiv:2003.10781]. (43 citations) 5. Gittins, F., Andersson, N., <i>Population synthesis of accreting neutron stars emitting gravitational waves</i>, Mon. Not. R. Astron. Soc. 488, 99 (2019) [arXiv:1811.00550]. (33 citations)	
SUBMITTED PUBLICATIONS	[28] Yu, H. et al., <i>Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions</i> [arXiv:2607.07943]. [27] Negri, L. et al., <i>Impact of the Einstein Telescope’s duty cycle on the estimation of binary black holes parameters</i> [arXiv:2606.19201]. [26] Gittins, F. et al., <i>Detecting Tidal Resonances in Binary Neutron Stars</i> [arXiv:2606.06376].	
COLLABORATION PUBLICATIONS	Since 2026, I have been a co-author on refereed LIGO-Virgo-KAGRA and Einstein Telescope Collaboration publications. I only list short-author-list publications below.	
REFEREED PUBLICATIONS	[25] Andersson, N., Counsell, A. R., Gittins, F., Ghosh, S., <i>Tidal response of a relativistic star</i>, Phys. Rev. D 113, 064051 (2026) [arXiv:2511.05139]. [24] Yin, S., Andersson, N., Gittins, F., <i>A post-Newtonian approach to neutron star oscillations</i>, Class. Quantum Gravity 42, 235002 (2025) [arXiv:2504.06918]. [23] Pnigouras, P., Andersson, N., Gittins, F., Counsell, A. R., <i>Dynamical neutron star tides: the signature of a mode resonance</i>, Mon. Not. R. Astron. Soc. 542, 1375 (2025) [arXiv:2508.06416]. [22] Counsell, A. R., Gittins, F. et al., <i>Interface modes in inspiralling neutron stars: A gravitational-wave probe of first-order phase transitions</i>, Phys. Rev. Lett. 135, 081402 (2025) [arXiv:2504.06181].	

- [21] **Gittins, F.**, Andersson, N., Yin, S., *Perturbation theory for post-Newtonian neutron stars*, *Class. Quantum Gravity* **42**, 135014 (2025) [[arXiv:2503.03345](#)].
- [20] **Gittins, F.**, Andersson, N., *Neutron-star seismology with realistic, finite-temperature nuclear matter*, *Phys. Rev. D* **111**, 083024 (2025) [[arXiv:2406.05177](#)].
- [19] **Gittins, F.** et al., *Problematic systematics in neutron-star merger simulations*, *Phys. Rev. D* **111**, 023049 (2025) [[arXiv:2409.13468](#)].
- [18] Counsell, A. R., **Gittins, F.** et al., *Neutron star g modes in the relativistic Cowling approximation*, *Mon. Not. R. Astron. Soc.* **536**, 1967 (2025) [[arXiv:2409.20178](#)].
- [17] Counsell, A. R., **Gittins, F.**, Andersson, N., *The impact of nuclear reactions on the neutron-star g-mode spectrum*, *Mon. Not. R. Astron. Soc.* **531**, 1721 (2024) [[arXiv:2310.13586](#)].
- [16] Pnigouras, P., **Gittins, F.**, et al., *The dynamical tides of spinning Newtonian stars*, *Mon. Not. R. Astron. Soc.* **527**, 8409 (2024) [[arXiv:2205.07577](#)].
- [15] Beri, A. et al., *AstroSat and NuSTAR observations of XTE J1739-285 during the 2019-2020 outburst*, *Mon. Not. R. Astron. Soc.* **521**, 5904 (2023) [[arXiv:2303.13085](#)].
- [14] **Gittins, F.** et al., *Modelling Neutron-Star Ocean Dynamics*, *Universe* **9**, 226 (2023) [[arXiv:2304.05413](#)].
- [13] **Gittins, F.**, Andersson, N., *The r-modes of slowly rotating, stratified neutron stars*, *Mon. Not. R. Astron. Soc.* **521**, 3043 (2023) [[arXiv:2212.04892](#)].
- [12] Andersson, N., **Gittins, F.**, *Formulating the r-mode Problem for Slowly Rotating Neutron Stars*, *Astrophys. J.* **945**, 139 (2023) [[arXiv:2212.04837](#)].
- [11] Andersson, N., **Gittins, F.** et al., *Building post-Newtonian neutron stars*, *Class. Quantum Gravity* **40**, 025016 (2023) [[arXiv:2209.05871](#)].
- [10] Riley, J. et al., *Rapid Stellar and Binary Population Synthesis with COMPAS*, *Astrophys. J. Suppl. Ser.* **258**, 34 (2022) [[arXiv:2109.10352](#)].
- [9] **Gittins, F.**, Andersson, N., *Modelling neutron star mountains in relativity*, *Mon. Not. R. Astron. Soc.* **507**, 116 (2021) [[arXiv:2105.06493](#)].
- [8] **Gittins, F.**, Andersson, N., Jones, D. I., *Modelling neutron star mountains*, *Mon. Not. R. Astron. Soc.* **500**, 5570 (2021) [[arXiv:2009.12794](#)].
- [7] **Gittins, F.**, Andersson, N., Pereira, J. P., *Tidal deformations of neutron stars with elastic crusts*, *Phys. Rev. D* **101**, 103025 (2020) [[arXiv:2003.05449](#)].
- [6] Pereira, J. P., Bejger, M., Andersson, N., **Gittins, F.**, *Tidal deformations of hybrid stars with sharp phase transitions and elastic crusts*, *Astrophys. J.* **895**, 28 (2020) [[arXiv:2003.10781](#)].
- [5] **Gittins, F.**, Andersson, N., *Population synthesis of accreting neutron stars emitting gravitational waves*, *Mon. Not. R. Astron. Soc.* **488**, 99 (2019) [[arXiv:1811.00550](#)].

WHITE PAPERS

- [4] Dietrich, T. et al., *ESO Expanding Horizon White Paper: Revealing the properties of matter at supranuclear densities with gravitational waves* [[arXiv:2512.16971](#)].

REVIEW ARTICLES

- [3] **Gittins, F.**, *Gravitational waves from neutron-star mountains*, *Class. Quantum Gravity* **41**, 043001 (2024) [[arXiv:2401.01670](#)].

SOFTWARE
ARTICLES

- [2] Riley, J. et al., *COMPAS: A rapid binary population synthesis suite*, **J. Open Source Softw.** **7**, 3838 (2022).

CONFERENCE
PROCEEDINGS

- [1] Thomas, A. Stevenson, E., **Gittins, F.** et al., *Galactic Archaeology with TESS: Prospects for Testing the Star Formation History in the Solar Neighbourhood*, **EPJ Web Conf.** **160**, 05006 (2017) [[arXiv:1610.08862](#)].